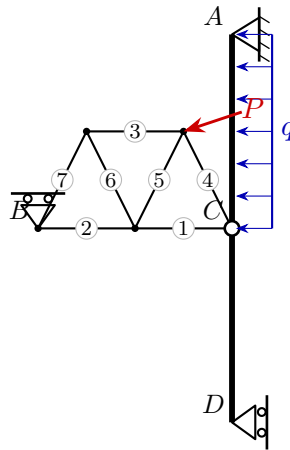


## Model solution



## 1. Static determinacy

Counting criterion  $n = 3k - (a + z)$  (bodies  $k$ , reactions  $a$ , interaction forces  $z$ ):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \Rightarrow \text{statically determinate}$$

## 2. Support reactions and hinge forces

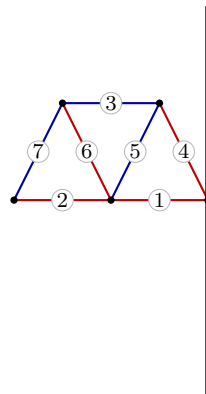
$$A: R_x = 14.25, R_y = -3.75 \quad D: R_x = 9.75, R_y = 0 \quad B: R_x = 0, R_y = 8.75$$

$$C: H_x = -15, H_y = 3.75$$

### 3. Truss member forces

| Member | Force $S$ [kN] | Type        |
|--------|----------------|-------------|
| 1      | 13.12          | tension     |
| 2      | 4.38           | tension     |
| 3      | -8.75          | compression |
| 4      | 4.19           | tension     |
| 5      | -9.78          | compression |
| 6      | 9.78           | tension     |
| 7      | -9.78          | compression |

red: tension (+)    blue: compression (–)



#### 4. Section-force diagrams of the beams

